QISKIT FALL FEST 2026 · ADVANCED TRACK

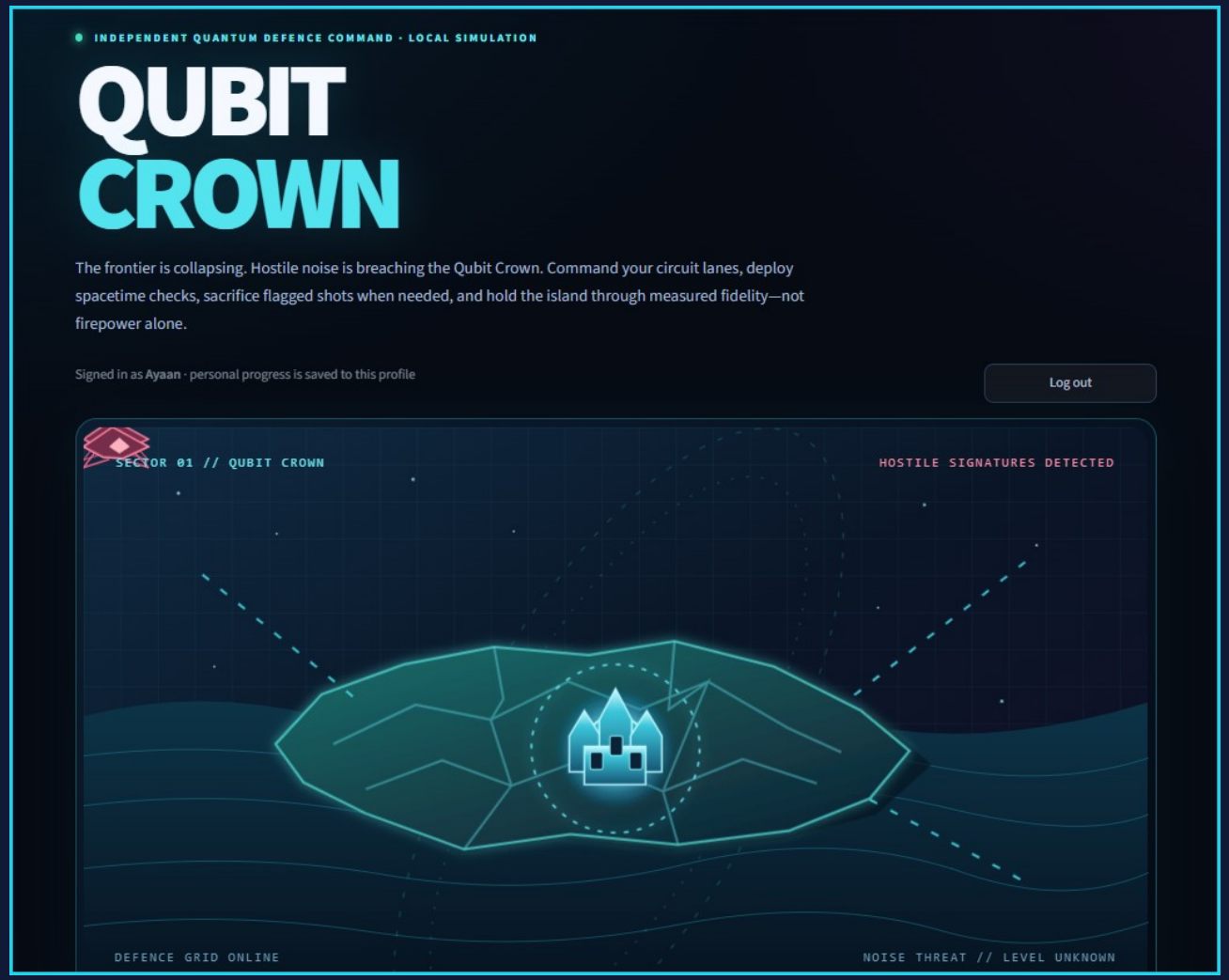
QUBIT CROWN

A strategy game for exploring quantum error detection

Team: **Ayaan and Vivaan**

Stack: **Qiskit · Qiskit Aer · qiskit-paulice**

Fidelity | Retention | Overhead



Live Qubit Crown main menu and Sector 01 tactical defense interface

Detecting errors comes at a cost.

CONCEPTUAL MODEL

Central Question: When does post-selection improve result quality enough to justify the shots and circuit resources it costs?

1. Noisy Circuit + Checks
Ancillas & parity gates add depth and 2Q gate noise before measurement.

→

2. Syndrome Filter
Ancilla outcomes classify each measurement shot as clean or flagged.

↗ **Accepted Shots (Kept):** Evaluated for Post-Selected Fidelity (F) · Yield R = N_acc / N_total

↘ **Rejected Shots (Discarded):** Thrown away — post-selection does NOT correct errors in flight.

1. Fidelity (F)

Result Quality

How closely the observed output distribution matches the ideal reference:

- Raw fidelity: before filtering.
- Post-selected: over accepted shots.
- Checks can lower fidelity if gate noise dominates.

2. Shot Retention (R)

R = accepted / total shots

The fraction of total measurement shots surviving post-selection:

- Every flagged shot is discarded.
- Stricter checks reject more runs, reducing usable sample yield.

3. Circuit Overhead

Added Hardware Cost

Extra circuit resources required to implement syndrome checks:

- Ancilla qubits added to the register.
- Extra two-qubit entangling gates.
- Deeper circuits and added decoherence.

GAMEPLAY LOOP

Turn error detection into a strategy.

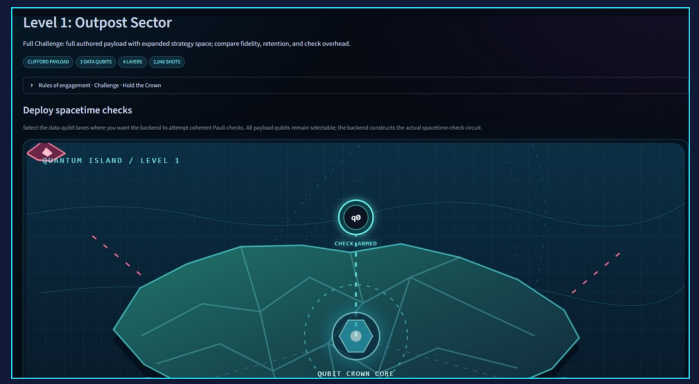
Players progress through five campaign sectors, allocate checks within a budget, and test strategies under noise.

STAGE 1 · CAMPAIGN & LEVEL SELECT



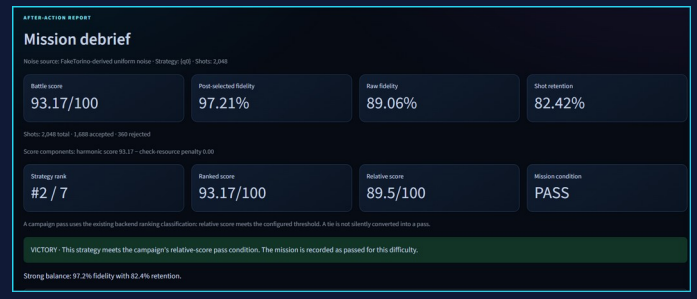
5 progressive sectors · 2,048 shots · Guided, Standard & Full Challenge tracks

STAGE 2 · TARGET QUBIT ALLOCATION



Level 1 Outpost Sector (3 data qubits, 4 layers) with q0 spacetime check armed

STAGE 3 · TELEMETRY & EVALUATION



Immediate debrief: 97.21% post-selected fidelity, 82.42% retention, PASS

01 · Choose Level

Select sector, difficulty preset & 2,048-shot config

02 · Select Qubits

Arm spacetime Pauli checks within check budget

03 · Simulate Shots

Execute noisy circuit runs in Qiskit Aer

04 · Inspect Metrics

Compare raw vs. post-selected fidelity & yield

05 · Rank Strategies

Evaluate composite score against all candidates

From noisy circuit to accepted results.

QISKIT + PAULICE

1. Payload Circuit
Clifford circuit
(3–7 data qubits)

→

2. Syndrome Checks
qiskit-paulice
spacetime checks

→

3. Noisy Simulation
Qiskit Aer
2,048 noisy shots

→

4. Syndrome Eval
Inspect ancilla
parity outcomes

→

5. Post-Selection
Filter flagged runs
keep clean shots

→

6. Compare Metrics
Fidelity, retention
& gate overhead

Technical Stack & Scope

▸ **Clifford Payload Circuits:** Campaign missions use Clifford-only payload circuits so stabilizer propagation remains exact.

▸ **qiskit-paulice Checks:** Constructs coherent spacetime Pauli checks (e.g., Check 1 support {q0, q1, q2, q3}) and ancilla registers (checks_q).

▸ **Qiskit Aer Simulation:** Executes 2,048 noisy shots under FakeTorino-derived uniform noise and records gate/depth overhead.

▸ **Detection, Not Correction:** Post-selection discards syndrome-flagged runs. It does NOT actively correct errors or perform fault-tolerant QEC.



Measured Level 1 {q0} Battle Report: +1 ancilla, 14 → 16 two-qubit gates, 20 → 29 depth, and Qiskit circuit diagram

Better accepted results. Fewer surviving shots.

LEVEL 1 · 2,048 SHOTS

NO-CHECK BASELINE

89.68%

Baseline Fidelity (unfiltered)

- **Retention:** 100% (2,048 kept)
- **Rejected:** 0 shots discarded
- **Circuit:** Depth 20 · 14 2Q · 0 anc
- **Score:** 91.98 / 100 (FAIL)

SYNDROME CHECK {q0}

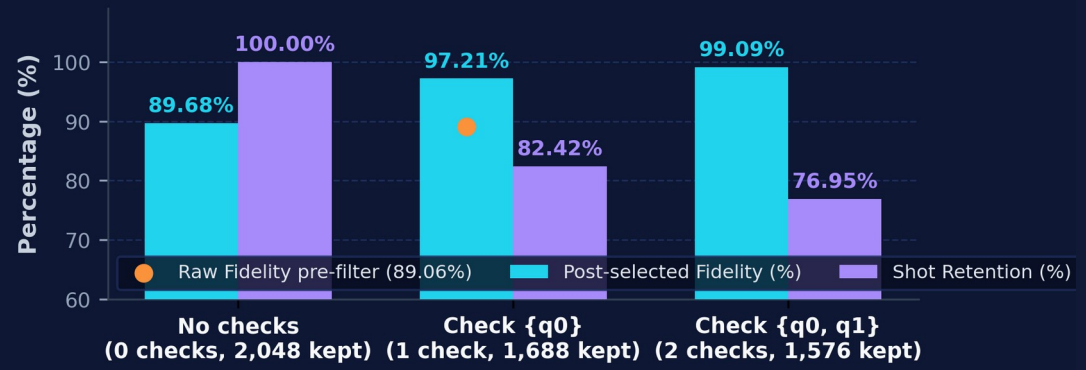
97.21% (+7.53 pp)

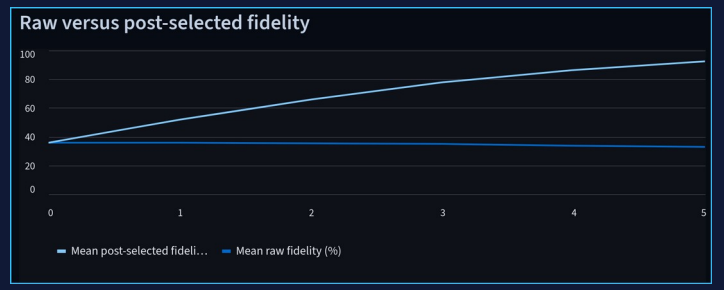
Post-Selected (Raw: 89.06%)

- **Retention: 82.42% (1,688 kept)**
- **Rejected:** 360 shots (17.6%)
- **Overhead:** Depth 29 · 16 2Q · 1 anc
- **Score: 93.17 / 100 (PASS #2)**



Actual Level 1 {q0} Mission Debrief: 97.21% post-selected vs. 89.06% raw fidelity, 82.42% retention





When Retention Collapses

In Level 5 (5 checks), post-selected fidelity climbs to ~94%, but retention drops to ~24% (+23 2Q gates, +96 depth) — proving higher fidelity alone does not make a strategy better.

Which strategy is worth the cost?

7 RANKED STRATEGIES

Strategy ranking for this mission

Highest-scoring strategy: {q2} · **93.31/100**

Score range: 91.98–93.31 · 7 valid strategies ranked. PASS threshold: relative score ≥ 50/100.

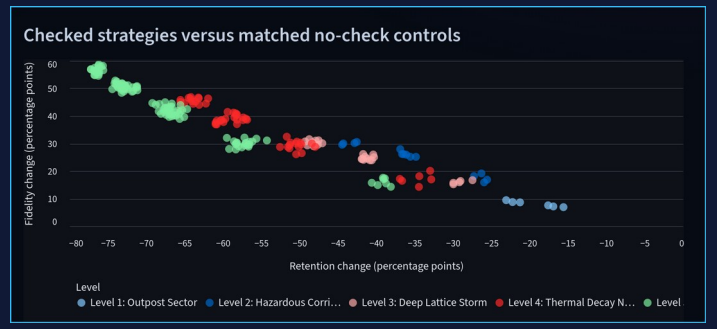
No-check baseline: 91.98/100. 6 of 6 checked strategies score higher.

Rank	Strategy	Selected	Checks	Score	vs baseline	Relative score	Outcome	Fidelity (%)	Retention (%)	Accepted shots	Highlights
1	{q2}		1	93.31	Better	100.0	PASS	96.55	84.42	1729	Max score
2	{q0}	YES	1	93.17	Better	89.5	PASS	97.21	82.42	1688	
3	{q1}		1	93.11	Better	85.0	PASS	96.83	83.11	1702	
4	{q0, q2}		2	92.75	Better	57.9	PASS	98.31	78.76	1613	
5	{q0, q1}		2	92.65	Better	50.4	PASS	99.09	76.95	1576	Max fidelity
6	{q1, q2}		2	92.48	Better	37.6	FAIL	98.41	77.78	1593	
7	No checks		0	91.98	Baseline	0.0	FAIL	89.68	100.00	2048	Max retention

Level 1 Leaderboard: {q0, q1} has max fidelity (99.09%, Rank #5), while 1-check {q2} wins max score (93.31, Rank #1)



Fidelity vs. shots kept by check count (0–5 checks)



Fidelity vs. retention shift across all 5 campaign levels

VERIFIED COMPOSITE SCORE

Balancing Quality vs. Yield

Ranked by a weighted harmonic score minus a separate check-cost penalty:

$$H_{\beta_l} = \frac{(1+\beta_l^2)F \cdot R}{F + \beta_l^2 R}$$

- **F (Post-selected fidelity):** Conditional quality of accepted shots.
- **R (Shot retention):** Fraction of total shots kept (N_acc / N_total).
- **β_l (Level parameter):** Configures the relative weight of retention vs. fidelity (e.g., $\beta_1 \approx 1.77$ in Level 1).
- **Separate Check Penalty:** Subtracted from H_β after the harmonic term (UI: harmonic score 93.17 − penalty 0.00).

Key Takeaway: No single check count is universally optimal; 2 checks maximize fidelity, but 1 check wins overall score.

What changes beyond Clifford circuits?

T-GATE SWEEP (0–12)

Probing the Clifford Boundary

1. Why the Campaign Uses Clifford Circuits

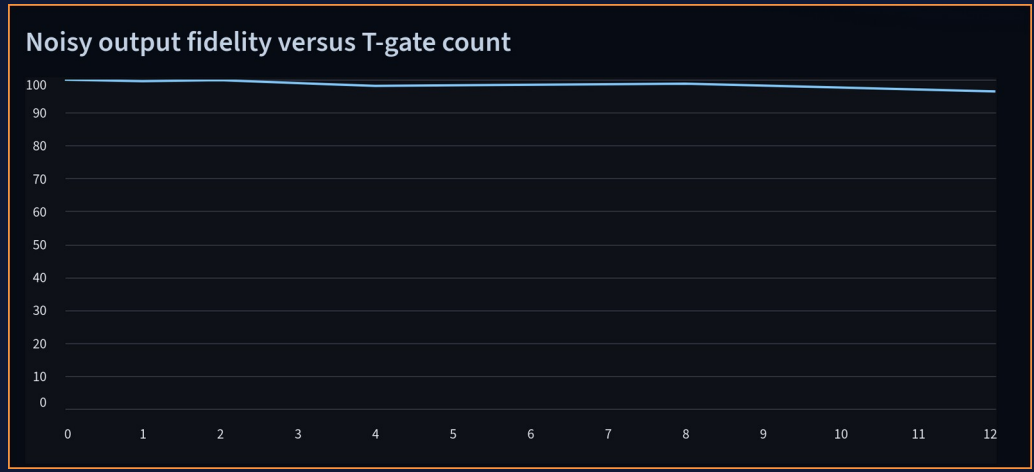
The campaign's qiskit-paulice syndrome-check workflow relies on stabilizer structure to propagate Pauli checks cleanly through Clifford gates.

2. The T-Doped Circuit Investigation

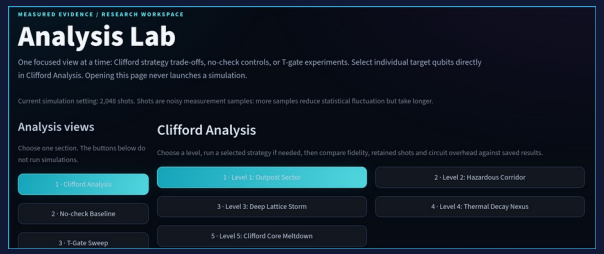
In Analysis Lab (3 · T-Gate Sweep), we separately investigate T/Tdg-doped circuits across 0 to 12 T-gates (48 logged experiments) to observe how non-Clifford gates behave under experimental noise.

3. Crucial Scientific Distinction

This T-gate sweep is a separate baseline investigation (Score: None; No checks). It is NOT proof that the Clifford syndrome-check workflow directly supports arbitrary non-Clifford circuits.



Measured Analysis Lab Sweep: Noisy output fidelity across 0 to 12 T-gates



Recorded Field Telemetry

- 48 T-Gate experiments logged in profile
- Dedicated '3 · T-Gate Sweep' workspace
- Logged 2,048-shot T-Gate baseline: 80.803% raw fidelity (100% retention)

Personal run history

When (UTC)	Type	Difficulty	Level	Strategy	Shots	Raw fidelity (%)	Post-selected fidelity (%)	Retention (%)	Score
2026-10-16T14:55:35+00:00	research	Full Challenge	Level 5: Clifford Core Meltdown	No checks	2048	35.791	35.791	100	42.9
2026-10-16T11:10:29+00:00	tgate	Full Challenge	T-Gate Research Experiment	No checks	2048	80.803	80.803	100	None

Error detection is a trade-off.

1. Post-selection changes the quality of accepted results.

Filtering syndrome-flagged runs raises accepted fidelity (e.g., 89.68% → 97.21% with {q0} in Level 1), evaluating conditional quality on surviving shots.

2. Rejected shots reduce the number of usable samples.

Flagged runs are discarded, not corrected in flight — cutting sample yield from 100% to 82.42% (1 check) and down to ~24% under heavy check loads.

3. Check circuitry introduces additional resource costs.

Ancilla qubits, extra two-qubit gates (14 → 16), and deeper circuits (20 → 29) add physical noise that must be weighed against fidelity gains.

Qubit Crown makes these trade-offs interactive and experimentally explorable.

Qubit Crown · Qiskit Fall Fest 2026 · Advanced Track · Team: Ayaan and Vivaan



Quantum Island · Level 5 (Clifford Core Meltdown): 7 data-qubit lanes around Qubit Crown Core